INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

PLOTTER, RECTANGULAR COORDINATES:

(COORDINATOGRAPH)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of precision plotters.

1.2 Classification. Plotters are of the following types, as specified (see 6.2):

Type I - Portable, with carrying case.
Type II - Table mounted (including light table).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: SATBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6675

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SPECIFICATIONS

FEDERAL

A-A-1898	- Cushioning Material, Packaging, (Cellulosic, Water Absorbent).
TT-I-528	- Ink, Drawing, Waterproof, Black.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-T-60	- Tape, Packaging, Waterproof.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproof, Waterproofed, Flexible.
MIL-P-514	- Plate, Identification, Instruction and Marking, Blank.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test methods.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951	- Standard Practice for Commercial Packaging.
D 3953	- Strapping, Flat Steel and Seals.
D 4675	- Selection and Use of Flat Strapping Materials.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The plotter shall consist of a movable ordinate member, stationary abscissa member, ordinate carriage unit and an abscissa carriage unit. The ordinate carriage unit, as a component of the ordinate member, shall have provisions for an insert holder and inserts.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.2.1 Material deterioration prevention and control. The plotter shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the plotter may be exposed.

3.2.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.2.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.2.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin

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raw materials. The components, pieces and parts incorporated in the plotter may be newly fabricated from recovered materials to the maximum extent practicable, provided the plotter produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the plotter.

3.3 Performance. The plotter drafting instrument shall draw pencil, pen or scribe lines which shall not vary by more than 0.05 millimeter (mm) from a taut, thin wire.

3.4 Scale ratios. The following scale ratios shall be furnished:

1 mm = 1 mm
1 inch = 1 inch
1 : 6,000 metric
1 : 10,000 metric
1 : 12,500 metric
1 : 15,000 metric
1 : 17,000 metric
1 : 25,000 metric
1 : 50,000 metric

3.5 Identification marking. The plotter shall be identified in accordance with MIL-STD-130. The marking shall be applied to the plotter on plates conforming to MIL-P-514, type I, style 3, composition C of type I, grade A, class 1 material. Each plate shall be attached by screws, bolts or rivets in a location where the plate will be both visible and legible.

3.6 Type I, portable, with carrying case. The type I plotter shall be capable of being disassembled for transporting and shall weigh not more than 84 pounds with carrying case and components. The instrument shall have a minimum working range of 650 mm in the ordinate "y" direction and 1000 mm in the abscissa "x" direction. The "y" axis shall be perpendicular to the "x" axis to within ±0.1 mm. The plotter shall be capable of plotting dimensionally related points to within ±0.1 mm.

3.6.1 Abscissa member. The abscissa member shall have sufficient weight to remain positioned on the drawing surface during operation of the ordinate member. The abscissa member shall have numbered scales engraved or permanently marked.

3.6.2 Abscissa carriage unit. The abscissa carriage unit shall be movable on the abscissa member and shall have a locking device, a magnifying lens with a tiltable mount for reading the abscissa vernier, and a vernier plate.

3.6.3 Ordinate member. The movable ordinate member shall have a large roller to support the free end and shall have numbered scales engraved or permanently marked. The ordinate member shall be mounted perpendicular to the

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abscissa member and shall be removable from the abscissa. The member, with its ordinate carriage unit, shall be freely movable over the entire design range of the plotter.

3.6.4 Ordinate carriage unit. The ordinate carriage unit shall be movable on the ordinate member and shall have a locking device, magnifying lens with a tiltable mount for reading the ordinate vernier, an insert holder and a vernier plate.

3.6.5 Insert holder. A holder (for inserts) shall be a part of the ordinate carriage unit and the following interchangeable inserts shall be furnished with the holder:

Pencil, lead (adjustable).
Pen, ruling (adjustable).
Point, pricking-scribing (adjustable).
lens, magnifying (with a circle-positioning reticle).

3.6.6 Scribing points. Each plotter shall be furnished with rigid, round scribing points of solid carbide or corrosion-resisting steel with corundum or tungsten carbide tips, as specified (see 6.2). Scribing points furnished shall be in tip sizes 0.004 inch 0.008 inch and 0.010 inch.

3.6.7 Carrying case. A carrying case shall be furnished for the plotter and the tools and accessories. The carrying case shall have provisions for carrying by two people. The case shall be labeled "TWO PERSON LIFT".

3.6.8 Plate, orienting. A plate shall be furnished which will permit adjusting the instrument to an existing plan.

3.6.9 Vernier plates. The verniers shall subdivide the scales.

3.7 Type II, table mounted (including light table). The type II plotter shall have a plotting range of 1200 mm in the ordinate direction and 1500 mm in the abscissa direction. The "y" axis shall be perpendicular to the "x" axis within ±0.05 mm. The plotter shall be capable of plotting dimensionally related points to within ±0.03 mm. Electrical components shall operate on 115-volt, single-phase, 60 Hertz. All electrical equipment and wiring shall conform to NFPA No. 70. A plate which states "115 volts, 60 Hz, single phase" shall be securely affixed near the electrical input.

3.7.1 Abscissa member. The abscissa member shall have unnumbered engraved or permanently marked scales which are compatible with interchangeable numbered tapes and vernier dials.

3.7.2 Abscissa carriage unit. The carriage unit shall be movable on the abscissa member and shall have a locking device and a vernier dial.

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3.7.3 Ordinate member. The movable ordinate member shall be mounted perpendicular to the abscissa member and shall have unnumbered engraved or permanently marked scales which are compatible with the interchangeable numbered tapes and vernier dials.

3.7.4 Ordinate carriage unit. The ordinate carriage unit shall be movable on the ordinate member and shall have a locking device, a vernier dial and a holder.

3.7.5 Table. A table for the plotter shall be furnished. The table shall accommodate the plotting range of the instrument. The table shall have a glass top covered with a translucent plastic sheet and shall be illuminated from the back by fluorescent lights. The lights shall be replaceable. The table shall be mounted on a frame supported by legs that incorporate screws for leveling the plotter.

3.7.6 Holder. A holder (for inserts) shall be a part of the ordinate carriage unit and the following interchangeable inserts shall be furnished with the holder:

Pencil, lead (with adjustable chuck).
Pen, ruling (adjustable and carbide tipped).
Point, pricking-scribing (with adjustable pricking depth).
Microscope 7X magnification field view and 3.5 mm diameter, with a circle (positioning reticle) and pricker.
Scriber point holder (spring loaded).

3.7.7 Points. The following scribing points shall be furnished with the plotter:

Tool Steel, single line.	- With cutting widths of 0.1 mm (0.004 in.), 0.2 mm, and 0.3 mm. (Tolerance: ±0.0127 mm.)
Diamond- or corundum-tipped, single line.	- with cutting widths of 0.08 mm, 0.1 mm, 0.2 mm, and 0.3 mm. (Tolerance: ±0.0127 mm.)

3.7.8 Tapes. The tapes furnished with the plotter shall be used on the abscissa and ordinate members, shall have reels, shall be numbered for scale ratios specified in 3.4; shall be interchangeable and shall have numbers advancing left to right.

3.7.9 Dials. The dials used on the abscissa and ordinate carriages with scale ratios conforming to 3.4 shall have a vernier and shall have 2-way numerations. The millimeter dial shall have a reading of 0.1 mm. The vernier used with the millimeter dial have a least reading of 0.01 mm.

3.7.10 Gage, orienting. A gage shall be furnished which shall permit adjusting the instrument to an existing plan.

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3.8 Human factors engineering. The plotter shall comply with the design criteria of MIL-STD-1472. Special design emphasis shall be given, but not limited to MIL-STD-1472, general requirements (4), control/display integration (5.1), visual displays (5.2), controls (5.4), labeling (5.5), design for Maintainer (5.9), and hazards and safety (5.13), as applicable.

3.9 Workmanship. All parts, components and accessories of the plotter, including castings, forgings, molded parts, stampings, bearings, machined surfaces and welded parts, shall be clean and free from sand, dirt, fins, sprues, scale, flux and other harmful extraneous material. All scale graduations engraved on the abscissa and ordinate members and verniers shall be legible. When a table is required, all surfaces shall be finished and preserved. The table frame and legs shall be rigidly constructed and joined to ensure stability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Material inspection. The contractor is responsible for ensuring that materials used are manufactured, examined and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3).
b. Inspection of packaging (see 4.5).

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4.3 Quality conformation inspection.

4.3.1 Examination. Each plotter shall be examined as specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Each plotter shall be tested as specified in 4.4.2.1 through 4.4.2.5, as applicable. Failure of any test shall be cause for rejection.

4.4 Inspection procedure.

4.4.1 Examination. The plotter shall be examined as specified herein for the following major defects:

101. Materials not as specified (see 3.2).
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.2.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.2.2).
104. Contractor does not have documentation available for identification of materials, material finishes or treatment (see 3.2.3).
105. Used, rebuilt or remanufactured components, pieces or parts incorporated in the plotter (see 3.2.4).
106. Scale ratios not as specified (see 3.4).
107. Identification marking missing, incomplete or illegible (see 3.5).
108. Abscissa member not as specified (see 3.6.1 and 3.7.1).
109. Abscissa carriage unit not as specified (see 3.6.2 and 3.7.2).
110. Ordinate member not as specified (see 3.6.3 and 3.7.3).
111. Ordinate carriage unit not as specified (see 3.6.4 and 3.7.4).
112. Insert holder missing or not as specified (see 3.6.5 and 3.7.6).
113. Scribing points missing or not as specified (see 3.6.6 and 3.7.7).
114. Carrying case not as specified (see 3.6.7).
115. Orienting plate not as specified (see 3.6.8).
116. Vernier plate not as specified (see 3.6.9).
117. Electrical components, equipment or wiring not as specified (see 3.6 and 3.7).
118. Table not as specified (see 3.7.5).
119. Tapes not as specified (see 3.7.8).
120. Dials not as specified (see 3.7.9).
121. Orienting gage missing or not as specified (see 3.7.10).
122. Human factors engineering not as specified (see 3.8).
123. Workmanship not as specified (see 3.9).

4.4.2 Tests.

4.4.2.1 Test conditions. Accuracy tests shall be performed in air-conditioned rooms maintained within ±2 °F of any temperature in the 60 to 75 °F range. Relative humidity (rh) shall not change more than 5 percent

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during measurements. Measurements and comparison checks shall be made immediately after plotting and drafting to minimize dimensional changes in the plastic drafting materials.

4.4.2.2 Drafting.

4.4.2.2.1 Pencil. Using a sharpened 3 H or 4 H pencil lead, draw gridlines 20 centimeters (cm) apart in the "x" direction across the entire length of a stable draft-coated polyester-base film having dimensions that approximate the plotters plotting range. Next draw gridlines in the "y" direction across the width of the drafting sheet. Check line straightness by using a 0.05- to 0.1- mm flexible wire weighted at each end with 1/4-pound (lb.) weights and stretched adjacent and parallel to the drawn line. Lines that are broken, ragged or vary in distance from the stretched wire more than 0.05 mm shall constitute failure of this test (see 3.3).

4.4.2.2.2 Pen. Using the pen provided with the plotter and using ink conforming to TT-I-528, draw gridlines 20 cm apart in the "x" direction across the entire length of the stable draft-coated polyester-base film. Next, draw gridlines in the "y" direction across the entire width of the drafting sheet. Line straightness shall be checked as in 4.4.2.2.1. Lines that are broken, ragged or vary in distance more than 0.05 mm from the stretched wire shall constitute failure of this test (see 3.3).

4.4.2.2.3 Scribing. Using the scribing point (0.1 mm or 0.004 inch) provided with the instrument, on stable draft-coated polyester-base film, scribe gridlines 20 cm apart in the "x" direction across the entire length of the scribe-coated sheet. Next, scribe gridlines in the "y" direction across the entire width of the sheet. Check line straightness as specified in 4.4.2.2.1. Lines that are broken, ragged or vary in distance more than 0.05 mm from the stretched wire shall constitute failure of this test (see 3.3).

4.4.2.3 Angle (90 degrees - "y" to "x").

4.4.2.3.1 Type I plotter. To determine if the ordinate carriage ("y" axis) is positioned at 90 degrees to the abscissa carriage ("x" axis), tape a sheet of stable polyester-base film approximately 26 inches by 40 inches to the table top. Select a scale on the abscissa member and aline the zero of this scale with the corresponding zero of the abscissa vernier plate. Place the ordinate carriage unit at zero. This position should be such that the pricking point provided with the plotter is approximately 2 inches from the lower left corner of the taped draft sheet (see figure 1, point "A"). Insert the pricking point into the holder located as a component of the ordinate carriage unit. Prick hole "A" into the draft sheet. After pricking the hole, raise the needle from the draft-sheet surface before moving the ordinate carriage unit. Loosen the locking device of the ordinate carriage unit and slide it slowly to the right until the pricking point is positioned approximately 2 inches from the right-hand edge of the draft sheet. Retighten the locking device. Prick hole "B". Leaving the ordinate carriage unit in

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this position, move the ordinate member slowly to the back edge of the draft sheet approximately 23 inches. Prick hole "C". Loosen the locking device. Leave the ordinate member in this position and move the ordinate carriage unit back to the starting zero and lock. Prick hole "D". Move the ordinate member from the test area. Using a magnifying glass and beam compass, adjust the compass-needle points until they exactly match the center-of-hole to center-of-hole distance on one diagonal of the rectangle ABCD. Move the beam compass to the other diagonal, place one compass point in a hole, and scribe a faint line near or through the related diagonal hole. A 40X shop microscope with measuring reticle shall be used to measure the distance from the scribed line to the center of the hole representing the end of the diagonal. A difference greater than ±0.1 mm shall constitute failure of this test (see 3.6).

4.4.2.3.2 Type II plotter. To determine if the ordinate member ("y" axis) is positioned at 90 degrees to the abscissa member ("x" axis), tape a sheet of stable polyester-base film approximately 42 by 42 inches to the table top. Select a scale printed on the numbered tape of the abscissa member and aline the zero mark of this scale with a graduation mark of the unnumbered scale on the abscissa member. This position should be such that the pricking point, provided with the plotter, is approximately 2 inches from the lower left corner of the taped draft sheet (see figure 1, point "A"). Select the vernier scale dial having the same scale ratio as the tape, insert it in the abscissa member and set the index mark to zero. Insert the pricking point into the holder located as a component of the ordinate carriage unit. Prick hole "A" into the draft sheet. After pricking the hole, raise the needle away from the draft-sheet surface before moving the ordinate carriage. Loosen the locking device of the ordinate carriage unit and slide it slowly to the right until the pricking point is positioned approximately 2 inches from the right-hand edge of the draft sheet. Retighten the locking device. Prick hole "B". Leaving the ordinate carriage unit in this position, move the ordinate member slowly to within 2 inches of the back edge of the draft sheet. Prick hole "C". Loosen the locking device. Leave the ordinate member in this position and move the ordinate carriage unit to the starting zero and lock. Prick hole "D". Move the ordinate member from the test area. Compare diagonal as specified in 4.4.2.3.1. A difference greater than ±0.05 mm shall constitute failure of this test (see 3.7).

4.4.2.4 Plotting. The invar meter bar used for this test shall be certified by the National Bureau of Standards as being capable of marking measurements with a probable error not greater than ±0.005 mm (see 3.6 and 3.7).

4.4.2.4.1 Type I instrument.

4.4.2.4.1.1 "X" axis. Place the certified invar meter bar parallel to the abscissa member so that the positioning circle of the microscope reticle, provided with the plotter, is split by the graduation line of the invar meter bar at the zero mark. Using the 1 mm - 1 mm abscissa scale, set the related vernier plate to zero to aline with the scale zero. Move the ordinate member along the abscissa member to 10-cm on the invar scale. Split the microscope

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reticle circle with the 10-cm graduation line and record the reading of the vernier plate. The 10-cm mark shall be set at least three times from both directions, and the average of not less than six vernier readings shall determine the exact abscissa measurement at that interval. Continue at 10-cm intervals to 100-cm. A difference greater than ±0.1 mm between any averaged measurement and the certified invar meter bar value at that point shall constitute failure of this test (see 3.6).

4.4.2.4.1.2 "Y" axis. Place the certified invar meter bar parallel to the ordinate member so that the positioning circle of the microscope reticle, provided with the plotter, is split by the graduation line of the certified invar meter bar at the zero mark. Using the 1 mm - 1 mm ordinate scale, set the ordinate vernier plate to zero to aline with the zero of the ordinate scale. Move the ordinate carriage unit along the ordinate member to the 10-cm mark on the invar scale. Split the microscope reticle circle with the invar meter bar graduation and record the reading of the vernier plate. The 10-cm mark shall be set at least three times from both directions, and the average of not less than six vernier readings shall determine the exact ordinate measurement at that interval. Continue at 10-cm intervals to 60-cm. A difference greater than ±0.1 mm between any averaged measurement and the certified invar meter bar value at the point shall constitute failure of this test (see 3.6).

4.4.2.4.2 Type II instrument.

4.4.2.4.2.1 "X" axis. Place the certified invar meter bar on the table top parallel to the abscissa member and positioned so that the circle of the microscope reticle, provided with the plotter, is split by the graduation line of the certified invar meter bar at the zero mark. Using the 1 mm - 1 mm abscissa scale, set the abscissa tape to zero at the first graduation of the unnumbered scale on the "x" axis. Aline the vernier dial to zero. Move the ordinate member so as to split the microscope reticle with the certified invar meter bar 10-cm mark. Record the deviation, if any exists, of the vernier dial from 10-cm. The 10-cm mark shall be set at least three times from both directions, and the average of not less than six vernier readings shall determine the exact abscissa measurement at that interval. Continue at 10-cm intervals to 100-cm. A difference greater than ±0.03 mm between any averaged measurement and the certified invar meter bar value at that point shall constitute failure of this test (see 3.7).

4.4.2.4.2.2 "Y" axis. Place the certified invar meter bar on the table parallel to the ordinate member so that the microscope reticle circle, provided with the plotter, is split by the zero graduation line of the meter bar. Using the same scale ratio as used for the "x" axis test, position the zero of the ordinate member tape to the first graduation on the unnumbered scale on the "y" axis. Aline the dial to zero. Move the ordinate carriage unit along the ordinate member to the 10-cm point on the invar scale. Record the deviation on the vernier dial from 10-cm. The 10-cm mark shall be set at least three times from both directions, and the average of not less than six vernier readings shall determine the exact ordinate measurement at that

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interval. Continue at 10-cm intervals to 100-cm. A difference greater than ±0.03 mm between any averaged measurement and the certified invar meter bar value at that point shall constitute failure of this test (see 3.7).

4.4.2.5 Type II scribing point cutting width. The tool steel, diamond- and corundum-point widths specified shall be tested by scribing lines on a sheet of stable polyester-base film. With a 40X shop microscope (as specified in 4.4.2.3.1), measure the scribed line widths of the specified points. Deviations of more than ±0.0127 mm from the specified widths shall constitute failure of this test (see 3.7.7).

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

124.	Components not cushioned and secured within the cases or boxes, as applicable, in a manner to prevent movement or damage for level A (see 5.1.1.2.1).
125.	Table not unassembled as specified for level A (see 5.1.1.2.1).
126.	Surfaces requiring a preservative not preserved and wrapped or covered as specified for level A (see 5.1.1.2.1).
127.	Working surface of the table not protected as specified for level A (see 5.1.1.2.1).
128.	Table not placed in box and contents not cushioned as specified for level A (see 5.2.1).
129.	Box closure not as specified for level A (see 5.2.1).
130.	More than one complete plotter packed together for level A or level B (see 5.2).
131.	Contents not secured within the shipping box in accordance with the referenced standard for level A or level B (see 5.2.1 and 5.2.2).
132.	Strapping not as specified for level A (see 5.2.1).
133.	Packing for commercial not as specified by the referenced document (see 5.2.3).
134.	Marking missing, illegible, incorrect or incomplete for level A, level B or commercial (see 5.3.1 and 5.3.2).

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Type I. The components of the type I plotter contained in the carrying case shall be preserved in accordance with MIL-P-116, method IIb. The carrying case shall be considered the inner container, and the outer container shall be a close-fitting box conforming to PPP-B-636, W5c, style optional.

5.1.1.2 Type II. The components comprising a complete type II plotter, except for the table (see 5.1.1.2.1), shall be contained in the contractor's cases or boxes, with the contents cushioned and secured to prevent movement or damage, and preserved in accordance with MIL-P-116, method IIb. The contractor's case or box shall be considered the inner container, and the outer container shall be a close-fitting box conforming to PPP-B-636, W5c, style optional.

5.1.1.2.1 Table. The table shall be unassembled and components requiring the application of a contact preservative in accordance with MIL-P-116, shall be lightly coated with type P-7 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116. Coated surfaces shall be covered or wrapped with barrier material conforming to MIL-B-121, grade A, type II, class 2, secured in place with tape conforming to PPP-T-60, type IV. The working surface of the table shall be covered with 1-inch-thick cushioning material conforming to A-A-1898, type II, class A, style and size optional and secured in place with tape specified herein. Each complete table shall then be placed in a close-fitting box conforming to PPP-B-636, V11c or V13c. The legs and any other loose components shall be wrapped with cushioning specified herein and shall be secured within the box in a manner to prevent movement or damage. Box closure shall be as specified for method V in the appendix to the box specification.

5.1.2 Commercial. The components comprising a complete plotter shall be preserved to afford protection against deterioration and damage during shipment and handling from the contractor to the initial destination.

5.2 Packing. Packing shall be level A, level B or commercial, as specified (see 6.2).

5.2.1 Level A. Each complete type I or type II plotter, preserved as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, grade B, style optional. The contents shall be secured within the box in a manner to prevent movement or damage in accordance with MIL-STD-1186. Box closure shall be in accordance with the appendix to the box specification. Strapping shall conform to ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

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5.2.2 Level B. Packing shall be as specified in 5.2.1 for level A, except the box shall be domestic type. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.3 Commercial. The components comprising a complete plotter shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Level A. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking shall be in accordance with ASTM D 3951. In addition, each shipping container shall display its gross weight and cube.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The plotter is used to prepare grid networks and plot dimensionally related points that are necessary for cartography.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number and date of the specification.
b. Type of plotter required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Scribing point material required (see 3.6.6).
e. Degree of preservation and degree of packing required (see 5.1 and 5.2).
f. When any special marking is required other than finish B, strapping is required (see 5.3).

6.3 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.4 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of plotters.

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6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Part or identifying number (PIN). The PIN to be used for plotter, rectangular coordinates (coordinatograph) acquired to this specification are created as follows:

M-52307-A

- A — Designator for type: A-type I, B-type II.
- 52307 — Specification number.
- M — Designator for document using inch-pound units.

6.7 Subject term key word listing.

Coordinates
Coordinatograph
Plotter

Custodians:
Army - ME
Air Force - 99

Review activity:
DLA-GS

Preparing activity:
Army - ME

Project 6675-0951

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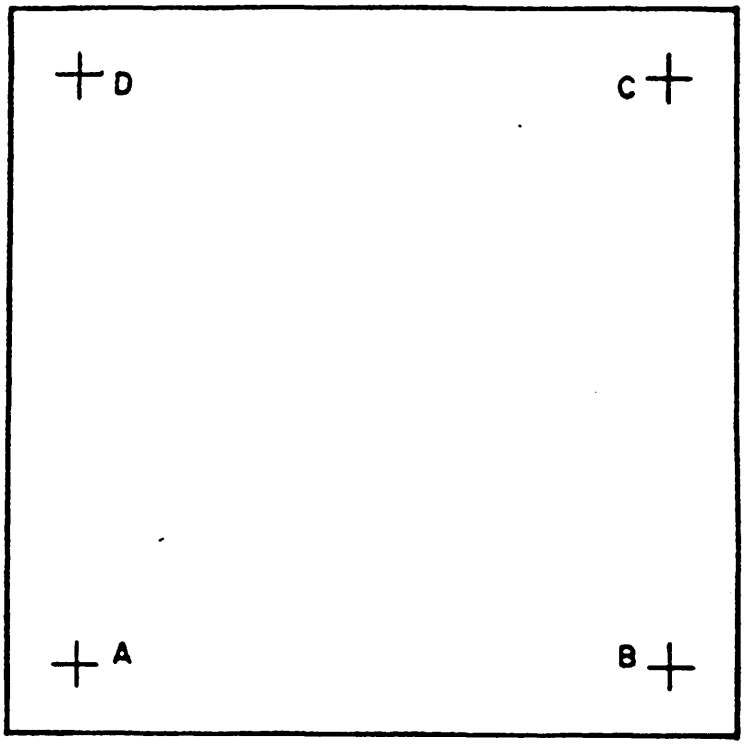


FIGURE 1. Dotting needle location(ABCD).

X-3224

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
   The submitter of this form must complete blocks 4, 5, 6, and 7.
   The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-P-52307B	2. DOCUMENT DATE (YYMMDD) 92/09/03

3. DOCUMENT TITLE
Plotter, Rectangular Coordinates: (Coordinatograph)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Violet Stewart	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3467 (2) AUTOVON 654-3467
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: SATBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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